UX-0008 — Butlerly Cross-Device Experience and Continuity
Version 1.0

Purpose
This document defines Butlerly's future optional cross-device experience across smartphone, tablet, and desktop. For Finance Version 1, single-device local-first operation is authoritative; trusted-device setup, synchronization, Apple ecosystem integration, AirDrop/manual transfer, cross-device conflict handling, and revocation are future-design references and are not MVP requirements.

1. Experience Objective
Butlerly should remain one product expressed through native smartphone, tablet, and desktop experiences. Finance Version 1 must work fully on a single device with a local database and no account. If optional cross-device continuity is implemented later, moving between devices should preserve meaning, progress, and trust without forcing users to understand synchronization internals.

The experience must:
- Put smartphone first for capture, quick review, and immediate action.
- Use tablet for richer review, comparison, and flexible work.
- Use desktop for bulk import, deep review, analysis, export, and administration.
- Preserve one shared information model, terminology, record identity, and history.
- Keep local work available when connectivity or providers fail.
- Keep single-device local operation complete. Future optional synchronization should be quiet when healthy and understandable when degraded.
- Prevent silent data loss.
- Preserve source evidence and earlier versions.
- Remain provider-independent if Apple or other synchronization services are introduced in a later release.

2. Governing Principles
2.1 Native Experiences, Shared Product
Layouts, navigation, input methods, and window behavior adapt to each device. Record meaning, user permissions, privacy rules, AI behavior, synchronization semantics, and recovery rules do not change by device.

2.2 Local First
Every trusted device maintains an encrypted local database. Local capture, search, review, editing, and history remain available without continuous connectivity.

2.3 Synchronization Is Transport, Not Ownership
A synchronization provider transports encrypted changes. It is not the authoritative owner of Butlerly data. The user and the trusted-device set remain authoritative.

2.4 Quiet When Healthy
Normal synchronization does not occupy permanent visual space. Status is available in My Devices and the Status Center. Delays, failures, security events, and unresolved integrity issues become visible.

2.5 Merge Before Conflict
Different records synchronize independently. Different fields of the same record merge automatically. A conflict exists only when competing changes cannot be reconciled without changing meaning.

2.6 Recovery Before Finality
Undo, version history, preserved source artifacts, and recoverable deletion are preferred over irreversible replacement.

2.7 Device Trust Is Explicit
A new device must be approved through an existing trusted device, a secure platform mechanism, or an encrypted recovery process. Device identity and revocation must remain visible.

3. Device Roles
3.1 Smartphone
Primary jobs:
- Capture receipts and documents.
- Share content into Butlerly.
- Review one exception at a time.
- Search and retrieve records quickly.
- Approve device pairing.
- Receive privacy-safe alerts.
- Continue recent work.

Design emphasis:
- One-handed actions.
- Fast launch and capture.
- Single-column focused flows.
- Minimal interruption.
- Strong offline behavior.

3.2 Tablet
Primary jobs:
- Review source and interpreted record side by side.
- Resolve exceptions and duplicates.
- Search, browse, and organize records.
- Use keyboard, pointer, touch, drag and drop.
- Continue work started on smartphone or desktop.

Design emphasis:
• Adaptive sidebar.
• List-detail layouts.
• Multi-column comparison.
• Compact and expanded width continuity.

3.3 Desktop
Primary jobs:
• Bulk import and mapping.
• Multi-record review.
• Detailed source inspection.
• Advanced search and filtering.
• Export, privacy, device, and diagnostic administration.
• Long-form Assistant work where supported.

Design emphasis:
• Persistent navigation.
• Multi-pane workspaces.
• Keyboard-first operation.
• Tables and batch workflows.
• Window and panel restoration.

4. Trusted-Device Model
A trusted device is a device holding valid Butlerly identity material and authorized encryption keys.

Each trusted-device entry must include:
• User-defined device name.
• Platform and device type.
• Trust date.
• Last successful connection.
• Last successful synchronization.
• Pending-change status.
• Key or credential status.
• Current trust state: Active, Offline, Delayed, Revoked, or Recovery Needed.

Trust rules:
• The first device creates or imports the Butlerly identity.
• New devices require explicit approval.
• Sensitive trust actions require device authentication.
• A revoked device cannot submit valid new changes.
• Revocation does not remove local data from a device that is physically inaccessible; the UI must state this honestly.

• Remaining trusted devices continue functioning after one device is revoked.

5. Device Addition Methods
5.1 Apple Synchronization
Preferred in the Apple ecosystem when enabled by the user.

The setup flow must explain:
• Each device keeps an encrypted local copy.
• Apple infrastructure transports synchronized data.
• Butlerly data remains encrypted according to the architecture standard.
• The user can disable synchronization without losing the current local copy.
• Availability depends on the user's Apple account and platform services.
• Manual transfer remains an alternative.

5.2 QR Pairing
An existing trusted device displays a short-lived QR code. The new device scans it and presents device identity for approval.

Required behavior:
• Short expiration.
• One-time use.
• Visible device name and type.
• Fingerprint or verification phrase.
• Approve and Reject.
• Clear expired, invalid, and already-used states.
• Safe cancellation on either device.

5.3 AirDrop or Encrypted Manual Transfer
A trusted device creates an encrypted transfer package. The package may be delivered through AirDrop, Files, removable storage, or another user-controlled path.

Required behavior:
• The package is encrypted and time-bounded where practical.
• The receiving device validates package integrity.
• Authentication is required before import.
• The source device shows what scope is included.
• Failed import does not damage existing local data.
• The package can be invalidated or safely discarded.

6. First Device Experience
The first-device flow must:
1. Create or import the Butlerly identity.
2. Explain local encrypted storage.
3. Offer device authentication.

4. Offer optional Apple synchronization.
5. Complete a first useful task before introducing advanced device management.
6. Show My Devices with the current device as Active.

Butlerly Finance Version 1 is designed for complete single-device operation. Synchronization is deferred and, if introduced later, must remain optional.

7. New Device Experience

The new-device flow must:

1. Explain that an existing Butlerly identity is being added.
2. Offer Apple Sync, QR Pairing, or Encrypted Transfer as available.
3. Authenticate the user.
4. Verify device identity.
5. Request approval where required.
6. Prepare the encrypted local database.
7. Transfer or synchronize selected data.
8. Validate integrity.
9. Show Device Ready.
10. Offer Continue Recent Work.

The user may safely interrupt initial synchronization. Progress resumes without duplicating records or losing existing changes.

8. Continuity Model

Continuity preserves user intent rather than cloning every pixel or transient state.

8.1 Continuity Units

Butlerly may preserve:

- Current record.
- Current review item.
- Search query and filters.
- Import batch and progress.
- Draft edit.
- Assistant context within approved scope.
- Selected settings area.
- Scroll position when useful.
- Unsaved local work when safe.

8.2 Continue Recent Work

Home may show a non-intrusive Continue Recent Work card when:

- A meaningful task was recently active on another trusted device.
- The state is synchronized and safe to resume.
- The task is not complete, deleted, revoked, or privacy-restricted.

The card shows:
• Task or record name.
• Origin device.
• Last activity time.
• Resume action.
• Dismiss action.

It must not expose sensitive details in notifications or lock-screen previews by default.

8.3 State That Does Not Need Synchronization
Purely local or transient UI state may remain device-specific:
• Window size and position.
• Panel width.
• Hover state.
• Temporary menu state.
• Camera state.
• Unsaved search typing not yet submitted.
• Platform-specific accessibility preferences unless the user explicitly enables preference synchronization.

9. Synchronization Unit and Change Model
Each change must identify:
• Record or entity ID.
• Field or relationship affected.
• Operation type.
• Device ID.
• Timestamp.
• Logical version or causal metadata.
• Prior value reference where required.
• User, AI, import, or system origin.
• Consent basis when external processing contributed.

Changes are synchronized incrementally. A whole database replacement is not the normal synchronization mechanism.

10. Merge and Conflict Rules
10.1 Different Records
Changes to different records never conflict. All valid changes synchronize independently.

10.2 Same Record, Different Fields
Changes merge automatically.

Example:
• iPhone changes Category.

• iPad changes Note.
• Result contains both changes.

10.3 Same Record, Same Field

The latest valid timestamp determines the current value in Version 1. The prior value remains in version history when meaningful.

Example:
• iPhone changes Category at 10:01.
• iPad changes Category at 10:03.
• The 10:03 value becomes current after validation.

10.4 Relationships and Collections

For additive relationships or collections, valid independent additions merge. A remove-versus-update decision uses operation timestamps and integrity rules. Source artifacts are never silently discarded.

10.5 Delete Versus Update

Deletion is represented as a recoverable tombstone, not immediate physical erasure.

Rules:
• A later valid update may restore or supersede a recoverable deletion according to product policy.
• Permanent deletion requires explicit confirmation and authentication.
• Related-record consequences are shown.
• Recently Deleted preserves recovery during the retention period.

10.6 Unresolved Conflict

A conflict is routed to Review only when automatic rules cannot preserve meaning or integrity.

The conflict view shows:
• Competing values.
• Devices.
• Timestamps.
• Source evidence.
• Butlerly recommendation.
• Choose, edit, merge, or defer actions.

11. Time and Timestamp Requirements

Timestamp-based resolution assumes trustworthy device time handling. The system should use platform-validated time, server or provider reference time when available, monotonic sequencing, or causal metadata to reduce clock-skew risk.

The UX must not expose unnecessary technical detail. When a resolution is shown, use understandable language such as:
“Category from Jack’s iPad became current because it was updated later. The earlier value remains in Version History.”

12. Synchronization States
12.1 Synced
No persistent banner. Status available on demand.

12.2 Pending Changes
Local work is saved. Changes are waiting to reach other devices. Local work continues.

12.3 Initial Sync
Show progress, current phase, safe interruption, and remaining local usability.

12.4 Delayed
Show last successful sync, affected devices, stale-data risk, Retry, and Manual Transfer.

12.5 Offline
Show that local work remains available and changes will synchronize later.

12.6 Provider Degraded
Show affected capability, preserved local work, provider status, and alternatives.

12.7 Merge Completed
Usually quiet. Optional history explains automatic merge.

12.8 Same-Field Resolution
Usually quiet. Activity history identifies the current and superseded values.

12.9 Unresolved Conflict
Visible in Inbox and affected record. Does not block unrelated records.

12.10 Revoked Source
Changes signed by revoked credentials are rejected and recorded as a security event.

13. Offline Experience
Offline is a normal state, not an exceptional mode.

Available offline where local data exists:
• Capture and preserve source.
• Manual entry.
• Browse and search local index.
• Edit records.

• Review locally available evidence.
• View history.
• Queue imports supported locally.
• Use local AI when available.
• Queue synchronization changes.

Potentially unavailable offline:
• External AI.
• Connected financial-source refresh.
• Cloud-drive import.
• New Apple synchronization setup.
• Remote device approval when no direct transfer path exists.

The interface must state what is available now and what will happen later.

14. Privacy and Encryption Communication
Primary UI language should focus on user meaning:
• "Stored encrypted on this device."
• "Waiting to sync to your other trusted devices."
• "Encrypted transfer package ready for AirDrop."
• "This device no longer has Butlerly access."

The interface must disclose:
• Data location.
• Synchronization provider when relevant.
• Trusted-device scope.
• External-processing boundaries.
• Export and deletion controls.

It must not imply that revocation remotely erases an inaccessible device unless that capability is technically verified.

15. Device Revocation
Revocation flow:
1. Open Device Detail.
2. Confirm device identity and last activity.
3. Explain consequences.
4. Authenticate.
5. Revoke keys or credentials.
6. Confirm remaining trusted devices are unaffected.
7. Record the security event.
8. Reject future changes from revoked credentials.

The user may rename, review, or revoke a device. Re-adding a revoked device requires a new trust flow.

16. Lost Device Experience
My Devices must provide a clear Lost Device path.

It should:
• Recommend revocation.
• Explain what revocation can and cannot do.
• Show last synchronization and pending-change uncertainty.
• Recommend platform-level lost-device protections where appropriate.
• Preserve audit history.
• Avoid alarmist language.

17. Recovery and Reinstallation
Supported recovery scenarios:
• Reinstall on an existing trusted device.
• Replace a device.
• Recover from another trusted device.
• Import an encrypted recovery package.
• Restore a supported backup.

Recovery must:
• Authenticate the user.
• Validate identity and integrity.
• Avoid creating duplicate identities.
• Preserve version history where available.
• State what data is recoverable and what may be missing.
• Never claim full recovery before validation completes.

18. Preference Synchronization
Data synchronization and preference synchronization are separate choices.

May synchronize when enabled:
• Locale-related product preferences.
• Default views.
• Learned AI preferences.
• Notification categories.
• Theme choice.

Remain device-specific by default:
• Biometric method.
• Window layout.
• Local storage settings.

• Camera settings.
• Platform accessibility settings.
• Notification permission granted by the operating system.

19. Notification Rules

Cross-device notifications must be actionable and privacy-safe.

Allowed examples:
• New device approved.
• Device revoked.
• Import completed with exception count.
• Synchronization requires attention.
• Review item ready.

Rules:
• Avoid displaying merchant, amount, or document content on lock screens by default.
• Do not notify every successful synchronization.
• Do not duplicate the same alert across all active devices unnecessarily.
• Security notifications remain visible until understood.
• Deep links restore the intended task when safe.

20. Accessibility

Cross-device continuity must support:
• Screen-reader announcement of synchronization and device states.
• Keyboard completion of pairing, review, revocation, and conflict resolution.
• Large-text reflow.
• Non-color state communication.
• Visible focus.
• Reduced motion.
• Accessible QR alternatives and verification phrases.
• Clear device names rather than opaque IDs alone.
• Logical reading order for comparisons.
• Durable recovery when a transient Undo expires.

21. Platform Adaptation

21.1 iPhone

• Prominent Capture.
• Device approval through focused sheet or full-screen flow.
• QR scanner and AirDrop entry.
• One-item conflict review.
• Privacy-safe notifications.

21.2 iPad

• List-detail My Devices.

• Side-by-side conflict or version comparison.
• Drag-and-drop encrypted package import where supported.
• Keyboard and pointer support.

21.3 Desktop
• Detailed device administration.
• Multi-column synchronization diagnostics.
• Bulk export and recovery-package management.
• Keyboard-first conflict resolution.
• Window restoration without synchronizing window geometry.

22. Error and Recovery Messages
Every synchronization error must state:
• What failed.
• What was preserved locally.
• Which devices may be stale.
• Whether retry is safe.
• What alternatives exist.

Preferred:
“Your changes are saved on this Mac but have not reached your other devices. Retry synchronization or create an encrypted transfer package.”

Avoid:
“Sync error 409.”
“Cloud failed.”
“Your data may be gone.”

23. Privacy-Preserving Analytics
Permitted events may include:
• Pairing method selected.
• Pairing completed or failed.
• Initial sync duration bucket.
• Sync delayed state.
• Manual transfer selected.
• Conflict type.
• Conflict resolved or deferred.
• Device revoked.
• Continue Recent Work opened or dismissed.

Analytics must not include record contents, financial values, device fingerprints, encryption keys, QR contents, transfer packages, or private search and Assistant content.

24. Quality Metrics

Key measures:
• Device-add completion rate.
• Median time to usable second device.
• Initial-sync success rate.
• Percentage of changes merged automatically.
• Same-field resolution rate.
• Unresolved-conflict rate.
• Data-loss incidents: target zero.
• Recovery success rate.
• Offline task completion rate.
• Device revocation completion rate.
• Continue Recent Work success rate.
• Accessibility task completion across supported devices.

25. Required Test Scenarios
• Smartphone edits one record while tablet edits another.
• Two devices edit different fields of the same record.
• Two devices edit the same field at different times.
• Clock skew or delayed delivery.
• Delete on one device and update on another.
• Offline capture followed by later sync.
• Provider outage during initial sync.
• Interrupted AirDrop transfer.
• Expired QR code.
• Rejected new-device approval.
• Revoked device attempts to submit changes.
• Reinstall and recovery.
• Large data set initial sync.
• Accessibility with screen reader, keyboard, and large text.
• Continue Recent Work after completion, deletion, or revocation.

26. Security UX Requirements
• Pairing codes are short-lived and one-time.
• Verification information is human-readable.
• Device authentication protects sensitive actions.
• Keys and secrets are never displayed 27. Version 1 Scope
P0 — Finance MVP
• One-device local-first operation with no Butlerly account required.
• Encrypted/local data protection according to approved architecture and platform capabilities.
• Offline capture, browsing, search, review, and editing where local data exists.
• Local export, backup/restore, deletion, and recovery according to Product priorities.
• Version history and preservation of source evidence where required by the Finance domain.

P1 — Release hardening and portability

• Backup/restore refinements and recovery usability.
• Portability and manual user-controlled transfer where approved by Product requirements.

P2 / Future optional cross-device capability
• Apple synchronization or other synchronization providers.
• My Devices and trusted-device management.
• Initial and incremental device synchronization.
• Cross-device change propagation, field-level merge, and same-field conflict resolution.
• Synchronization status, delayed/provider-failure states, and device revocation.
• QR pairing, AirDrop/encrypted transfer for device pairing, and Continue Recent Work across devices.
• Cross-platform provider support, selective synchronization, and multi-device activity history.

Rule: none of the P2/future capabilities may introduce a mandatory Butlerly account or weaken single-device local operation.

lict resolution.
• Recovery package.
• Rich synchronization diagnostics.

P2:
• Cross-platform Android and Windows providers.
• Advanced selective synchronization.
• Family or delegated access.
• Rich multi-device activity timeline.

28. Dependencies and Governance
UX-0008 depends on:
• UX-0001 — Butlerly Experience Blueprint.
• UX-0002 — Butlerly Version 1 User Journey Map.
• UX-0003 — Butlerly Information Architecture and Navigation Model.
• UX-0004 — Butlerly Core User Flows.
• UX-0005 — Butlerly Screen and State Inventory.
• UX-0006 — Butlerly AI Interaction and Explainability Specification.
• UX-0007 — Butlerly Design System Foundations.
• Architecture decisions for identity, encryption, local database, synchronization, clocks, keys, provider abstraction, backup, and recovery.

UX-0008 governs:
• UX-0009 — Accessibility Standard.
• UX-0010 — Low-Fidelity Wireframe Specification.
• UX-0011 — Design Patterns and Component Library.
• Device, synchronization, conflict, recovery, notification, and continuity engineering stories.
• Cross-device design QA and release acceptance.

29. Acceptance Criteria

This specification is complete when:

- Smartphone, tablet, and desktop roles are explicit.
- Future trusted-device setup methods are documented as optional design references and are not required for Finance V1 completion.
- Local-first and offline behavior are defined.
- Different-record, different-field, same-field, deletion, and relationship merge rules are defined.
- Timestamp use is limited to competing updates and includes clock-integrity considerations.
- Finance V1 recovery and local-data behavior are represented; Apple Sync, QR pairing, AirDrop/device transfer, and revocation remain future optional references.
- Single-device local operation is complete without synchronization; future optional sync behavior remains defined for later use.
- Continuity preserves user intent without synchronizing unnecessary transient UI state.
- Privacy and encryption boundaries are understandable.
- Accessibility and analytics requirements are testable.
- Engineering can derive data-change metadata, state machines, and acceptance tests.
- QA can verify that Finance V1 local changes are preserved reliably; cross-device change validation applies only when synchronization is implemented later.

BDS-0000 Conformance

BDS-0000 — Butler Design System v1.0 governs all concrete visual tokens, typography, spacing/radius, responsive breakpoints, component/state behavior, accessibility implementation, localization/original-data presentation, and Flutter UI constraints used by any future cross-device capability. Finance V1 visible primary navigation remains Home, Transactions, Review, Search, and Settings. This document remains a future/optional continuity specification; it does not make account creation, device trust, pairing, or synchronization a core MVP dependency. UI-authored text follows the selected application language, while user-entered/imported/scanned/source data preserves its original language/script and original currency/amount by default.